2026 NASA SPACE APPS CHALLENGE · THE NEXT FRONTIER

# PROHOR

## Mission Health Intelligence

## Your readings. Your baseline. A clearer next step.

An offline-capable health workspace for astronauts who must assess themselves,
with no clinic and a delayed link to the ground.

**LIVE PROTOTYPE** prohor-rho.vercel.app **SOURCE** github.com/tufayelahmedsuhagcc-dotcom/Proho

Team: [names] · Challenge: Create Health Monitoring Software for Astronauts on Space Missions

Research prototype. Not a medical device. Independent project; no NASA endorsement implied.

# Up to twenty-two minutes of delay. Five readings. No clinic.

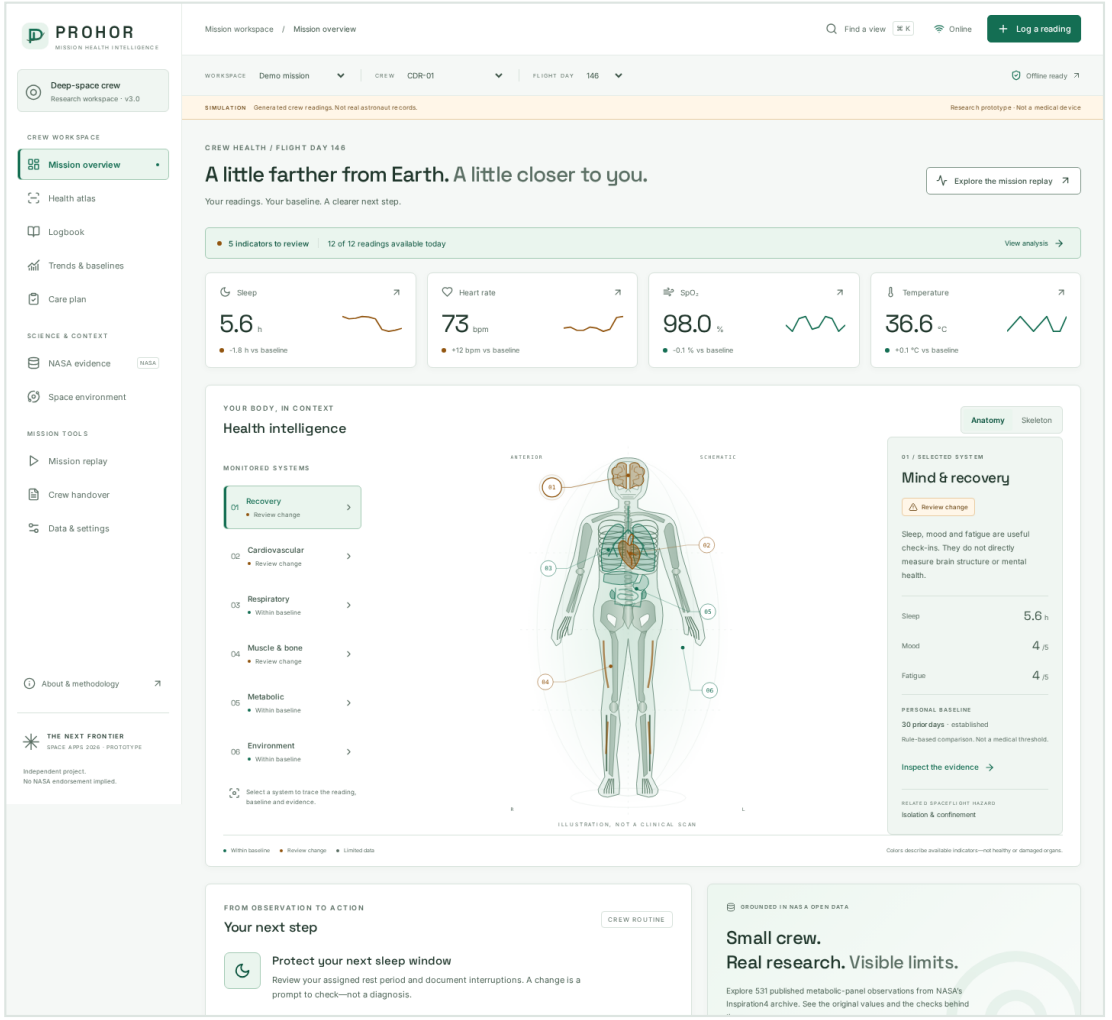
A crew member on a long-duration mission has to decide, alone, whether a number matters — and what to do next.

### What did I measure?
A validated entry, in its original units.

### How does it compare to my normal?
A baseline from this person's own prior days only.

### How sure is this?
Coverage, direction and the rule, shown with the result.

### What do I do now?
One conservative step, documented and handed to a surgeon.




Screenshot: mission overview, flight day 146 — five indicators under review.

02 / THE GAP

# A colour is not a reason.

## Most health dashboards answer with red, amber or green.

That answer hides the four things a crew member needs to judge it.

### A flag with no baseline

Was this compared with the crew member, or with a textbook number?

### A flag with no sample size

Two days of data is not a baseline. The interface rarely says which.

### A flag with no rule

A threshold cannot be checked, argued with, or corrected.

### A flag with no next step

An alarm without an action moves the decision back to the crew, unaided.

PROHOR's design rule: publish the uncertainty instead of a verdict.

# Comparison you can audit.

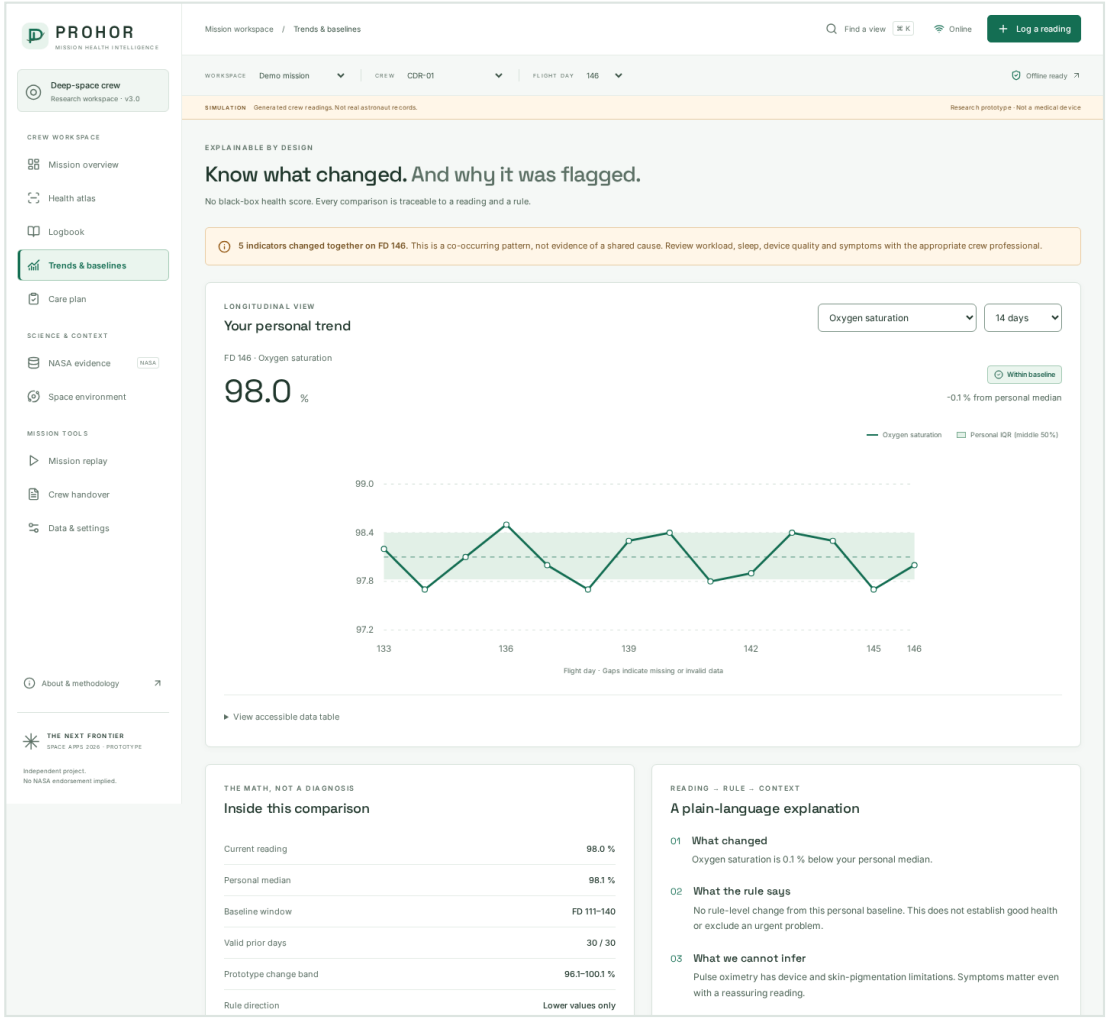
Median and interquartile range over the last 30 distinct prior days, excluding today, future days, invalid readings and missing values.

**7 days minimum**
below that, the app says so instead of reassuring you.

**21+ days**
labelled established — never a confidence score.

**Never borrowed**
no baseline from another crew member or another mode.

**Every flag opens**
window, day count, direction, source readings, table view.




Screenshot: trends & baselines — the rule, the band and the coverage on one screen.

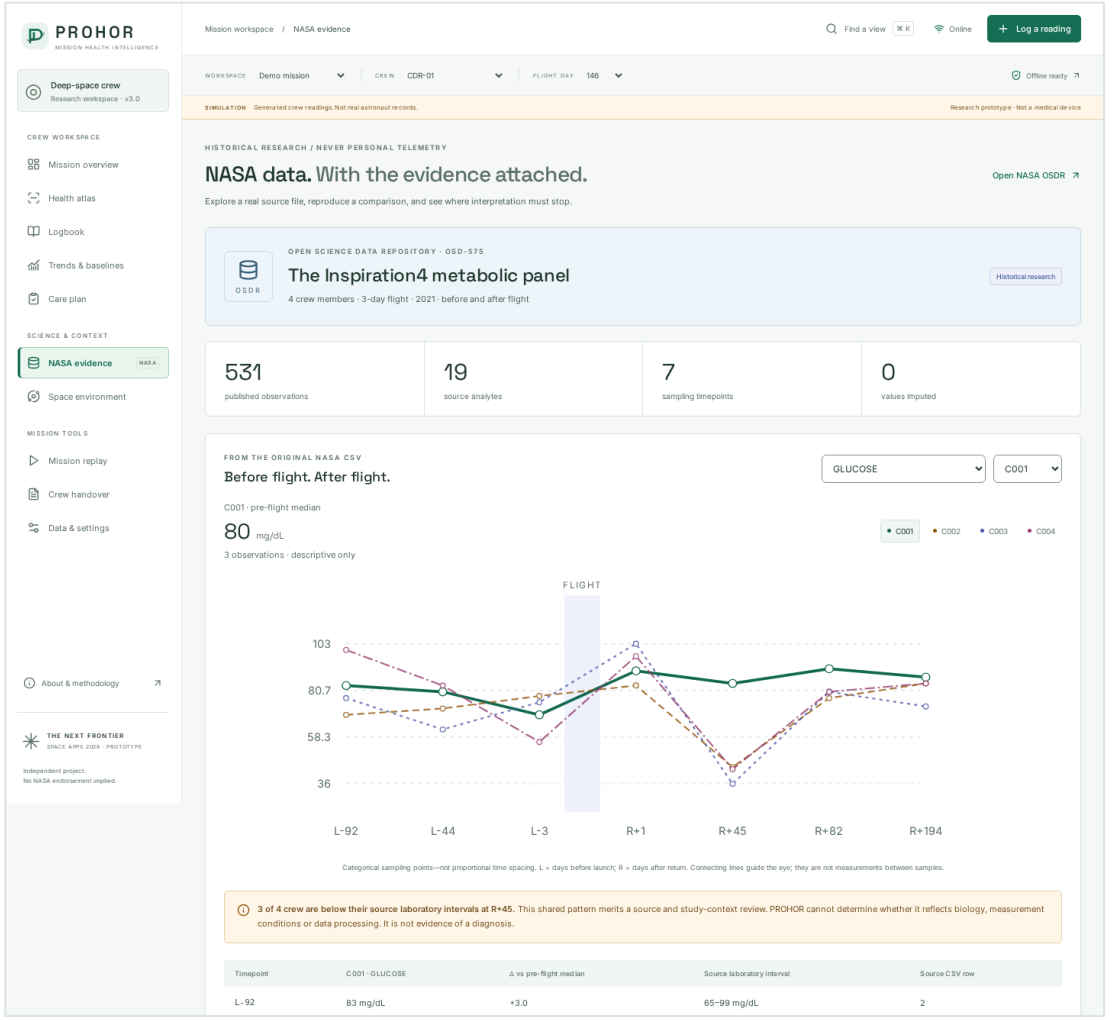
# NASA data, checked in front of you.

**OSD-575**
531 published Inspiration4 metabolic-panel observations, parsed in-browser.

**SHA-256**
the original CSV bytes are hashed and verified on load.

**Recomputed**
medians recalculated per analyte and subject, with source row numbers.

**Stated limits**
four crew, one short flight, shared sample intervals — said plainly.




Screenshot: NASA evidence — glucose, subject C001, pre-flight median 80 mg/dL.

# The packet, not the alert.

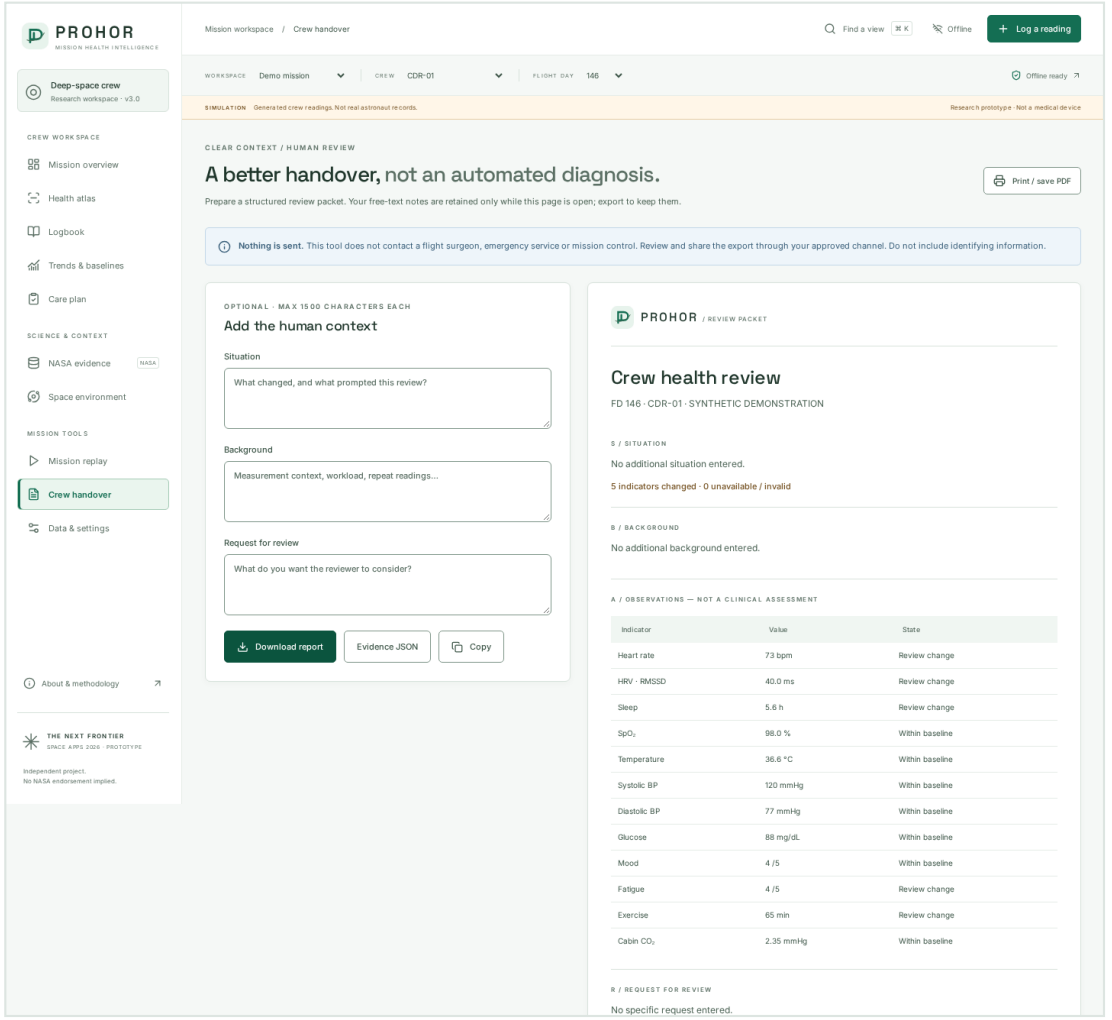
A flag ends in one conservative routine step, documented, and a structured handover the flight surgeon can read without re-deriving anything.

**Care plan**
done or deferred, persisted per crew, day and mode.

**Handover**
observations, baseline context, rule version, notes.

**Export**
text, JSON, clipboard or print-to-PDF. Nothing is transmitted.

**Offline**
installs from the browser and keeps working with no link.




Screenshot: crew handover — the review packet a surgeon receives.

# What this is not.

Stated because it protects the crew it is designed for:

**Not a medical device**
no diagnosis, treatment or dosing, no dose estimate, no fitness call.

**Not encrypted**
single-user browser storage; no sync, accounts or multi-crew roles.

**Not long-mission proof**
the archive is four crew on one short flight.

**NEXT**

- Indicator set reviewed with flight surgeons
- On-orbit measurement ingestion
- Multi-crew roles and audited access
- Accessibility and clinical review

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**SOURCE CODE** github.com/tufayelahmedsuhagcc-dotcom/Proho

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